A

INTERNSHIP REPORT ON

"Full Stack Web Application using MERN Stack"

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE

OF

THIRD YEAR OF ENGINEERING

(COMPUTER ENGINEERING)

SUBMITTED BY

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Area of Internship:- Full Stack Web Application using MERN Stack

Sem:- VIth

Sinhgad Institutes

DEPARTMENT OF COMPUTER ENGINEERING

STES'S SINHGAD ACADEMY OF ENGINEERING

KONDHWA, PUNE 411048

UNIVERSITY OF PUNE

2025-26

Sinhgad Institutes

DEPARTMENT OF COMPUTER ENGINEERING

STES'S SINHGAD ACADEMY OF ENGINEERING
KONDHWA, PUNE 411048

CERTIFICATE

This is to certify that the Internship report title

"Full Stack Web Application using MERN Stack"

Submitted by

Mr. PAWAR SIDDHESH SURYAKANT Exam No: T400430419

is a bonafide work carried out by him under the supervision of Prof. A. H. Kagne and it is submitted towards the Partial fulfillment of the requirement of Savitribai Phule Pune University, for the award of the degree of Third year of Engineering (Computer Engineering).

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It is indeed with a great sense of pleasure and immense sense of gratitude that I acknowledge the help of these individuals

Working in the **(CODEC TECHNOLOGIES)** was interesting. During these one months of internship, I learnt a lot web development, especially on completing three task all are helpful for understanding all concepts in HTML, CSS and JavaScript.

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(Student Name & Signature)

ABSTRACT

In This report presents the details of my internship training in MERN Stack Development at CODEC Technologies. The MERN stack consists of MongoDB, Express.js, React.js, and Node.js, which together form a powerful framework for building dynamic web applications.

During this training, I gained knowledge of both frontend and backend technologies. I learned how to design responsive user interfaces using React and how to manage server-side operations using Node.js and Express.js. Additionally, I worked with MongoDB for database management.

The training also included practical sessions where I developed real-world projects. These projects helped me understand how to integrate frontend and backend components effectively. I also learned about RESTful APIs and how to handle HTTP requests.

Version control tools like Git were introduced, enabling efficient code management. The internship emphasized hands-on learning, problem-solving, and debugging techniques. Overall, this training enhanced my technical skills and prepared me for real-world software development challenges. It provided a strong foundation in full-stack development and improved my ability to build scalable web applications.

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CODEC TECHNOLOGIES

NATIONAL INTERNSHIP PORTAL

INTERNSHIP OFFER LETTER

Dear SIDDHESH SURYAKANT PAWAR,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : MERN Stack Developer Intern
- Location : Hybrid / India
- Duration : 01/01/2026 to 30/01/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by Project Heads.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.
Best regards,

Sincerely,
Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: EI9E86-0116588288923
Date Of Issue - 01/01/2026

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INTERNSHIP COMPLETION CERTIFICATE

CODEC TECHNOLOGIES

NATIONAL INTERNSHIP PORTAL

CERTIFICATE OF INTERNSHIP

This certificate awarded to

Siddhesh Suryakant Pawar

From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC Approved internship program as

MERN Stack Developer Intern

Conducted from 01/01/2026 to 30/01/2026

Google for Education Partner

AICTE ID - CORPORATE6759d549ce59e1733940555

Dr. Anurag Shrivastava

Program Manager

INTERNSHIP PLACE DETAILS: COMPANY BACKGROUND

The internship was carried out at CODEC Technologies, a reputed institute known for providing high-quality training in the field of information technology and software development. The organization focuses on delivering industry-oriented education and bridging the gap between academic knowledge and practical skills.

CODEC Technologies offers a wide range of courses, including web development, full-stack development, programming languages, data science, and emerging technologies. The institute is committed to equipping students with the latest technical skills required in today's competitive IT industry. Their training programs are designed in alignment with current industry standards and practices.

The organization has a team of experienced trainers who possess strong technical expertise and real-world industry experience. They provide in-depth knowledge through interactive sessions, practical demonstrations, and real-time project guidance. The teaching methodology focuses on "learning by doing," which helps students understand concepts more effectively.

The infrastructure at CODEC Technologies is well-equipped with modern computer systems, high-speed internet connectivity, and updated software tools. The learning environment is professional and encourages students to actively participate in coding activities and discussions.

During the internship, the institute provided structured training in MERN Stack Development, covering MongoDB, Express.js, React.js, and Node.js. The program included theoretical sessions along with hands-on project work, enabling students to gain real-world development experience.

CODEC Technologies also emphasizes soft skills and career development. Students are guided in resume building, interview preparation, and problem-solving techniques. This holistic approach ensures that trainees are not only technically skilled but also professionally prepared.

The organization's mission is to develop skilled professionals who can contribute effectively to the IT industry. Its vision is to become a leading training provider by continuously updating its curriculum and adopting modern teaching techniques.

Overall, the internship at CODEC Technologies provided a supportive and motivating environment that enhanced my technical knowledge, practical skills, and confidence in software development.

INTRODUCTION

The MERN stack is a popular technology stack used for developing modern web applications. It includes MongoDB, Express.js, React.js, and Node.js. These technologies work together to provide a complete solution for frontend and backend development. During my internship at CODEC Technologies, I was introduced to the fundamentals of web development.

I learned how different components of a web application interact with each other. The training started with basic concepts such as HTML, CSS, and JavaScript. Gradually, it progressed to advanced topics like React components, API integration, and database management. The internship provided practical exposure through project development. This helped me understand how to implement theoretical concepts in real-world scenarios.

The main objective of this training was to build a strong foundation in full-stack development. It also aimed to improve problem-solving and coding skills. Overall, the introduction to MERN stack has opened new opportunities for me in the field of software development.

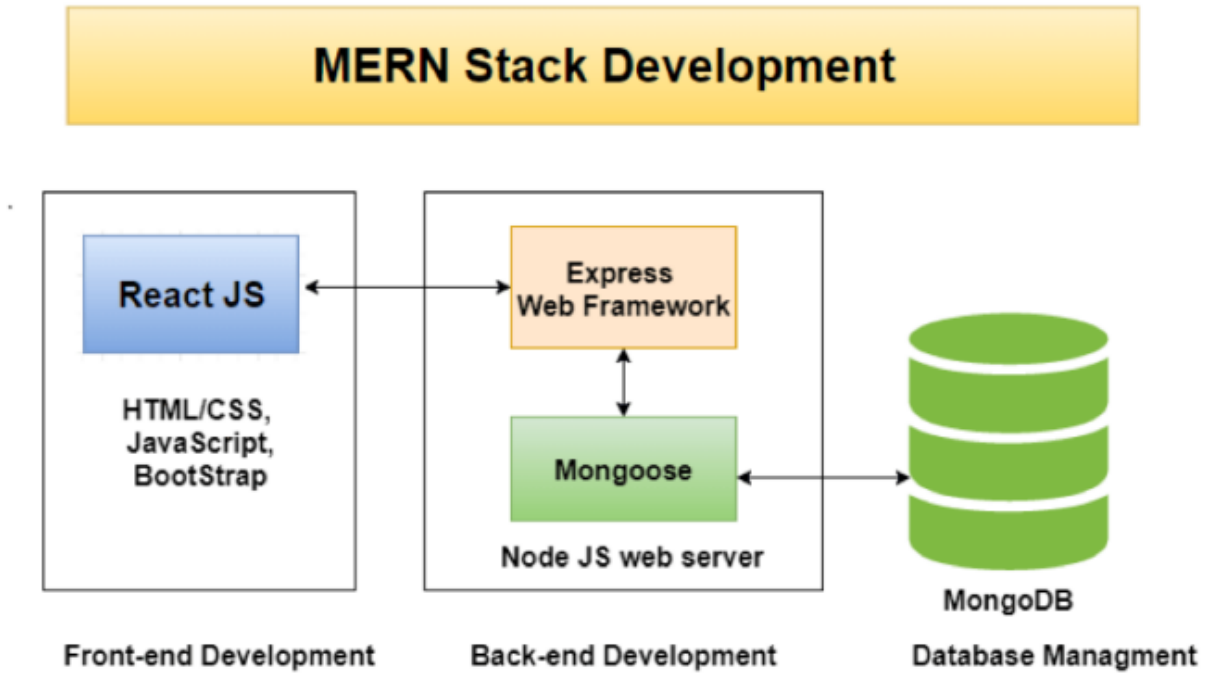

Fig. 1. Architecture

PROBLEM STATEMENT

The title of my internship project is “Full Stack Web Application using MERN Stack.” The main objective was to develop a dynamic and responsive web application. The problem statement involved creating a system that can handle user data efficiently and provide a smooth user experience.

The application needed to support CRUD operations (Create, Read, Update, Delete). Another challenge was integrating frontend and backend components seamlessly. Ensuring data security and proper API handling was also important. The project required implementing authentication and authorization features.

This ensures that only authorized users can access certain functionalities. The system needed to be scalable and efficient. Proper database design was essential to achieve this goal. Overall, the problem statement focused on building a complete web application using modern technologies. The problem statement involved creating a system capable of managing user data efficiently while providing a seamless user experience. The application needed to support CRUD operations and ensure data integrity.

One of the major challenges was integrating different technologies into a single application. This required a clear understanding of both frontend and backend processes. Security was another important aspect, requiring the implementation of authentication and authorization mechanisms.

The application also needed to be scalable and efficient to handle multiple users. Overall, the project focused on solving real-world problems using modern web technologies.

OBJECTIVES

The main objective of this internship was to learn MERN stack development. It aimed to provide knowledge of both frontend and backend technologies.

- Another objective was to develop practical skills through real-world projects. This helped in understanding how web applications are built.
- The training also focused on improving coding and debugging skills. Learning version control using Git was another important objective.
- Understanding REST APIs and database management was also a key goal. The internship aimed to enhance problem-solving abilities.
- It also focused on building responsive and user-friendly interfaces.
  Overall, the objectives were designed to prepare students for professional roles in software development.
- It aimed to provide hands-on experience in building web applications.
- Another objective was to understand the interaction between frontend and backend systems.
- The training focused on improving coding, debugging, and problem-solving skills.
- Learning database management and API development was also a key goal.
- The internship aimed to prepare students for real-world software development roles.
- It also encouraged teamwork and collaborative learning.

MOTIVATION

- The motivation behind learning MERN stack development is the increasing demand for full-stack developers. Companies require professionals who can handle both frontend and backend tasks.

- This internship provided an opportunity to gain such skills. The scope of MERN stack is vast and includes web and mobile application development.

- Learning these technologies opens doors to various career opportunities. It also allows developers to build scalable and efficient applications.

- The training motivated me to explore new technologies and improve my skills.

- The scope also includes working on real-time projects and contributing to open-source platforms.

- Overall, MERN stack development offers great career growth and opportunities.

- The motivation behind learning MERN stack lies in its high demand in the software industry.

- Full-stack developers are highly valued for their ability to handle complete projects.

- This internship provided an opportunity to learn these in-demand skills.

- The scope of MERN stack includes developing web applications, mobile apps, and enterprise solutions.

- It also opens opportunities in freelancing and startup development.

- The training motivated me to continuously learn and upgrade my skills.

- Overall, it provides a strong foundation for a successful career in IT.

DESIGN

The design phase involved planning the structure of the web application. It included creating wireframes and UI layouts.

React.js was used for designing the frontend interface. Components were created for different sections of the application.

CSS and Bootstrap were used for styling and responsiveness.

The backend design included creating APIs using Express.js. Node.js handled server-side logic.

MongoDB was used for database design and data storage.
Proper design ensured smooth functionality and user experience.

The design of the system focuses on simplicity, efficiency, and scalability. It is capable of handling multiple users simultaneously without affecting performance. The modular structure of the system allows easy maintenance and future enhancements.

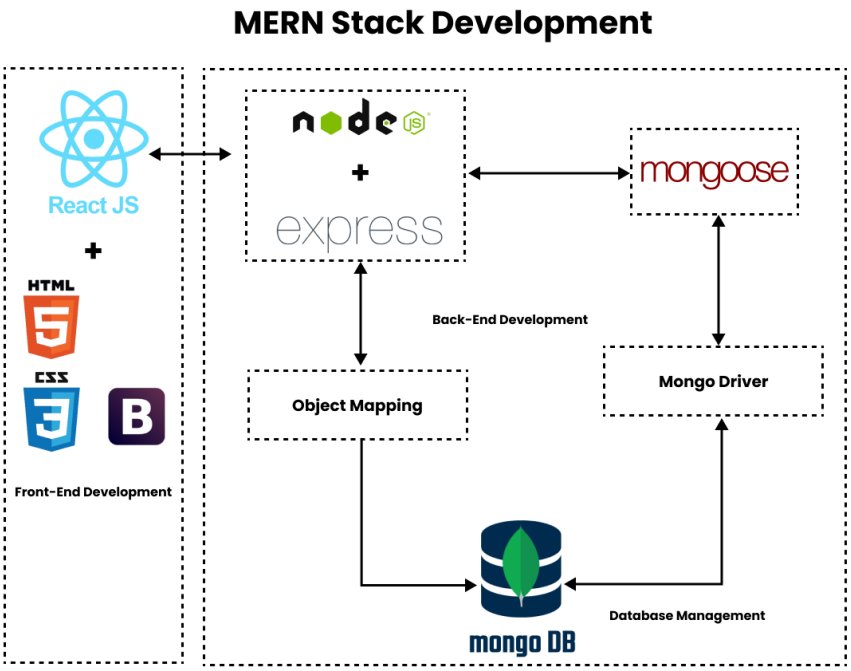

Fig. 2. Design

METHODOLOGIES USED

- The project followed an agile methodology. Development was done in small phases.
- Each phase included planning, development, testing, and deployment.
- Version control was maintained using Git.
- RESTful APIs were used for communication between frontend and backend.
- Debugging and testing were done regularly.
- This methodology ensured efficient and organized development

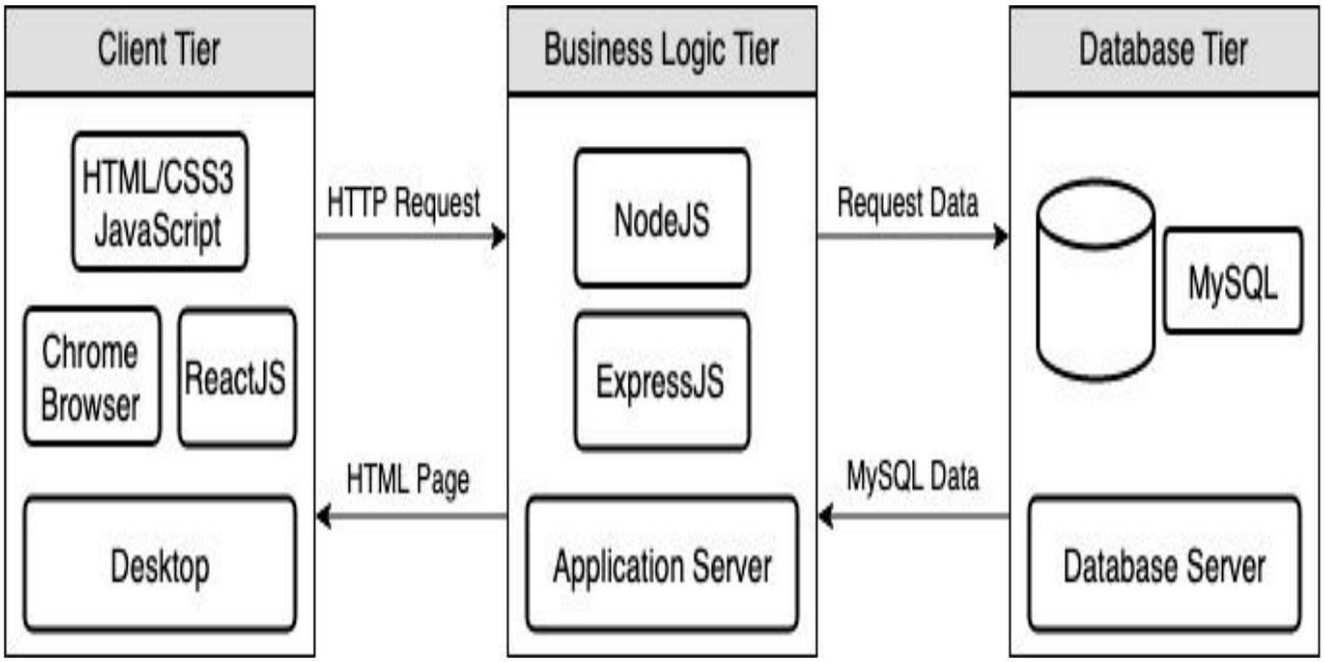

Fig. 3. System Workflow

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HARDWARE AND SOFTWARE USED

The development and implementation of the proposed e-waste management system require a combination of hardware components and software technologies to ensure smooth functioning, efficient processing, and user accessibility. Both hardware and software play a crucial role in building, testing, and deploying the application in a reliable environment.

Hardware Requirements

The hardware used for the development of this project primarily includes a computer or laptop with sufficient processing capability to handle development tools and runtime environments. A system with a minimum of 4 GB RAM and a modern processor is required to ensure smooth execution of applications such as code editors, browsers, and backend servers. Higher configurations further improve performance, especially during multitasking and testing phases.

A stable internet connection is another essential requirement, as it is necessary for accessing online resources, APIs, libraries, and tools used during development. It also enables integration with external services such as map services and chatbot APIs. Additionally, internet connectivity is required for testing the application in a real-time environment and for deployment purposes.

Input and output devices such as a keyboard, mouse, and display monitor are used for coding, debugging, and interacting with the system. These basic hardware components support efficient development and usability of the application.

Software Requirements

The software components used in this project are selected based on their efficiency, scalability, and compatibility with modern web application development.

The frontend of the application is developed using React.js, which allows the creation of dynamic and interactive user interfaces. It supports component-based architecture, enabling better code organization and reusability. CSS is used along with React.js to design and style the interface, ensuring that the application is visually appealing and responsive across different devices.

For backend development, Spring Boot is used as the primary framework. It simplifies the development process by providing built-in features such as configuration management, security, and rapid application setup. It enables the creation of RESTful APIs, which are essential for communication between the frontend and backend components of the system.

The system uses **MongoDB**, a NoSQL document-oriented database, to efficiently store and manage application data. MongoDB is chosen due to its flexibility, scalability, and ability to handle large volumes of unstructured and semi-structured data.

The system is developed using the **MERN stack**, which consists of MongoDB, Express.js, React.js, and Node.js. This stack is widely used for building modern, dynamic, and scalable web applications.

MongoDB is used as the database to store and manage application data in a flexible, JSON-like format. It allows efficient handling of large datasets and supports easy scalability.

Express.js is used as the backend web framework that handles server-side logic, API creation, and request-response management. It simplifies the development of robust and secure server-side applications.

React.js is used for building the frontend user interface. It enables the creation of interactive and responsive UI components, improving user experience and application performance.

Node.js acts as the runtime environment that allows JavaScript to be executed on the server side. It ensures fast performance and efficient handling of multiple requests through its non-blocking architecture.

The MERN stack provides seamless integration between frontend and backend, faster development using a single programming language (JavaScript), and improved performance. It is highly suitable for developing scalable and maintainable web-based systems.

RESULTS

- The project was successfully completed with all required features.
- The application performed CRUD operations efficiently.
- User interface was responsive and user-friendly.
- Backend APIs worked smoothly.

The project was successfully completed with all the required features and functionalities. The application efficiently performed CRUD operations, ensuring smooth data management throughout the system. The user interface was designed to be responsive and user-friendly, providing a better experience across different devices.

The backend APIs functioned effectively, enabling seamless communication between the frontend and the database.

Through this project, strong full-stack development skills were demonstrated using modern technologies. By the end of the internship, a complete web application was developed successfully, which helped in gaining confidence in working with MERN stack technologies.

Additionally, coding efficiency and logical thinking abilities were significantly improved. Overall, the project met all the defined objectives and the internship played an important role in enhancing both technical and professional skills.

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FUTURE SCOPE

The project can be further enhanced by incorporating advanced features to improve its overall functionality and performance. The authentication system can be strengthened by implementing secure methods such as JWT-based authentication. Additionally, integrating a cloud database and deploying the application on platforms like Heroku can improve accessibility and scalability.

Performance optimization techniques can be applied to ensure faster and more efficient operation of the system. Furthermore, features like real-time updates using WebSockets can be added to enhance user interaction. Improving the UI/UX design can also provide a better and more engaging user experience. Overall, these enhancements can make the application more scalable, efficient, and suitable for real-world use.

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CONCLUSION

The internship at CODEC Technologies was a great learning experience and provided valuable exposure to modern web development technologies. It helped in understanding MERN stack development in detail and gave a clear insight into how real-world applications are built. This experience played an important role in strengthening the foundation of full-stack development.

During the internship, the gap between theoretical knowledge and practical implementation was effectively bridged. Hands-on experience was gained by working on real-world applications, which significantly improved coding skills, problem-solving ability, and logical thinking. The training also enhanced technical knowledge and boosted confidence in working with development tools and technologies.

Overall, this internship proved to be a valuable step toward a career in software development. It not only improved technical skills but also prepared for real-world challenges in the industry. The experience gained during this period has contributed to both professional growth and future career opportunities.

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